

Local instance MySQL80 x

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

SCHEMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

Administration

Schemas

Information

No object selected

Bhartesh*SQL File 2*SQL File 4*SQL File 6* x

Don't Limit

3

4-- QUESTION 1:Display all records from the Superstore sales table.

5• SELECT *

6FROM SAMPLESUPERSTORE;

7

Result Grid

Filter Rows:

Export:

Wrap Cell Contents:

Fetch rows:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
▶	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	261.96	2	0	41.9136
	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.94	3	0	219.582
	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Labels	14.62	2	0	6.8714
	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	957.5775	5	0.45	-383.031
	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies	Storage	22.368	2	0.2	2.5164
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Furnishings	48.86	7	0	14.1694
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Art	7.28	4	0	1.9656
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	907.152	6	0.2	90.7152
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Binders	18.504	3	0.2	5.7825
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Appliances	114.9	5	0	34.47
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Tables	1706.184	9	0.2	85.3092
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	911.424	4	0.2	68.3568
	Standard Class	Consumer	United States	Concord	North C...	28027	South	Office Supplies	Paper	15.552	3	0.2	5.4432
	Standard Class	Consumer	United States	Seattle	Washing...	98103	West	Office Supplies	Binders	407.976	3	0.2	132.5922
	Standard Class	Home Off...	United States	Fort Worth	Texas	76106	Central	Office Supplies	Appliances	68.81	5	0.8	-123.858
	Standard Class	Home Off...	United States	Fort Worth	Texas	76106	Central	Office Supplies	Binders	2.544	3	0.8	-3.816

UPERSTORE 1 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 12	23:12:00	USE SQL_PROJECT	0 row(s) affected	0.000 sec
✓ 13	23:13:31	SELECT * FROM SAMPLESUPERSTORE	9994 row(s) returned	0.000 sec / 0.063 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Filter objects

SCHMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

SQL File 6* x samplesuperstore - Table

Don't Limit

```
-- Question 2: Display only the Category, Sub Category, Sales, Quantity, Discount, and Profit columns.
SELECT CATEGORY, "SUB CATEGORY", SALES, QUANTITY, DISCOUNT, PROFIT
FROM SAMPLESUPERSTORE;
```

Result Grid

	CATEGORY	SUB CATEGORY	SALES	QUANTITY	DISCOUNT	PROFIT
▶	Furniture	SUB CATEGORY	261.96	2	0	41.9136
	Furniture	SUB CATEGORY	731.94	3	0	219.582
	Office Supplies	SUB CATEGORY	14.62	2	0	6.8714
	Furniture	SUB CATEGORY	957.5775	5	0.45	-383.031
	Office Supplies	SUB CATEGORY	22.368	2	0.2	2.5164
	Furniture	SUB CATEGORY	48.86	7	0	14.1694
	Office Supplies	SUB CATEGORY	7.28	4	0	1.9656
	Technology	SUB CATEGORY	907.152	6	0.2	90.7152
	Office Supplies	SUB CATEGORY	18.504	3	0.2	5.7825
	Office Supplies	SUB CATEGORY	114.9	5	0	34.47
	Furniture	SUB CATEGORY	1706.184	9	0.2	85.3092
	Technology	SUB CATEGORY	911.424	4	0.2	68.3568
	Office Supplies	SUB CATEGORY	15.552	3	0.2	5.4432
	Office Supplies	SUB CATEGORY	407.976	3	0.2	132.5922
	Office Supplies	SUB CATEGORY	68.81	5	0.8	-123.858
	Office Supplies	SUB CATEGORY	2.544	3	0.8	-3.816

Result + x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	23:19:15	SELECT CATEGORY,SUB CATEGORY,SALES,QUANTITY,DISCOUNT,PROFIT FROM SA...	Error Code: 1054. Unknown column 'SUB' in 'field list'	0.000 sec
19	23:19:30	SELECT CATEGORY,"SUB CATEGORY",SALES,QUANTITY,DISCOUNT,PROFIT FROM ...	9994 row(s) returned	0.000 sec / 0.032 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: bhartesh* SQL File 2* SQL File 4* SQL File 6* samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

SCHEMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

Result Grid

Filter Rows:

Export: Wrap Cell Content: 12

CATEGORY	SUBCATEGORY	REGION
Furniture	Bookcases	South
Furniture	Chairs	South
Office Supplies	Labels	West
Furniture	Tables	South
Office Supplies	Storage	South
Furniture	Furnishings	West
Office Supplies	Art	West
Technology	Phones	West
Office Supplies	Binders	West
Office Supplies	Appliances	West
Furniture	Tables	West
Office Supplies	Paper	South
Office Supplies	Appliances	Central
Office Supplies	Binders	Central
Office Supplies	Storage	Central
Office Supplies	Storage	West
Office Supplies	Art	Central

SAMPLESUPERSTORE 14 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
32	23:29:38	SELECT DISTINCT CATEGORY,SUBCATEGORY,REGION FROM SAMPLESUPERSTORE	68 row(s) returned	0.062 sec / 0.000 sec
33	23:30:14	SELECT DISTINCT CATEGORY,SUBCATEGORY,REGION FROM SAMPLESUPERSTORE	68 row(s) returned	0.079 sec / 0.000 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

SQL File 2* SQL File 4* SQL File 6* x samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

Don't Limit

15

16 -- Question 4:Find all sales transactions where the Sales amount is greater than 500.

17 • SELECT *

18 FROM SAMPLESUPERSTORE

19 WHERE SALES>500;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Subcategory	Sales	Quantity	Discount	Profit
▶	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.94	3	0	219.582
	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	957.5775	5	0.45	-383.031
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	907.152	6	0.2	90.7152
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Tables	1706.184	9	0.2	85.3092
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	911.424	4	0.2	68.3568
	Standard Class	Consumer	United States	Madison	Wisconsin	53711	Central	Office Supplies	Storage	665.88	6	0	13.3176
	Standard Class	Consumer	United States	Orem	Utah	84057	West	Furniture	Tables	1044.63	3	0	240.2649
	Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furniture	Bookcases	3083.43	7	0.5	-1665.0522
	First Class	Corporate	United States	Richardson	Texas	75080	Central	Technology	Phones	1097.544	7	0.2	123.4737
	Standard Class	Home Office	United States	Houston	Texas	77041	Central	Furniture	Bookcases	532.3992	3	0.32	-46.9764
	Standard Class	Corporate	United States	New York City	New York	10024	East	Technology	Phones	1029.95	5	0	298.6855
	Standard Class	Corporate	United States	Gilbert	Arizona	85234	West	Office Supplies	Art	1113.024	8	0.2	111.3024
	Standard Class	Consumer	United States	Memphis	Tennessee	38109	South	Furniture	Chairs	831.936	8	0.2	-114.3912
	Standard Class	Consumer	United States	Seattle	Washington	98103	West	Furniture	Tables	787.53	3	0	165.3813
	Second Class	Home Office	United States	Houston	Texas	77041	Central	Furniture	Chairs	600.558	3	0.3	-8.5794
	Standard Class	Consumer	United States	Bloomington	Illinois	61701	Central	Furniture	Tables	617.7	6	0.5	-407.682

SAMPLESUPERSTORE 16 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✖ 35	23:33:06	WHERE SALES>500	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
✔ 36	23:33:49	SELECT * FROM SAMPLESUPERSTORE WHERE SALES>500	1162 row(s) returned	0.015 sec / 0.016 sec

No object selected

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

Bhartesh*

SQL File 2*

SQL File 4*

SQL File 6*

samplesuperstore - Table

samplesuperstore - Table

samplesuperstore - Table

Don't Limit

SQL Editor

25

-- Filtering, Sorting & Calculations - 6 to 10

26

-- Question 6:Display all transactions from the Consumer segment and sort them by Sales in descending order.

27

• SELECT * FROM SAMPLESUPERSTORE

28

WHERE SEGMENT='CONSUMER'

29

ORDER BY SALES DESC;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Subcategory	Sales	Quantity	Discount	Profit
▶	First Class	Consumer	United States	Seattle	Washington	98115	West	Technology	Copiers	13999.96	4	0	6719.9808
	Standard Class	Consumer	United States	Newark	Delaware	19711	East	Technology	Copiers	10499.97	3	0	5039.9856
	Standard Class	Consumer	United States	Detroit	Michigan	48205	Central	Office Supplies	Binders	9892.74	13	0	4946.37
	Standard Class	Consumer	United States	Minneapolis	Minnesota	55407	Central	Office Supplies	Binders	9449.95	5	0	4630.4755
	Second Class	Consumer	United States	Arlington	Virginia	22204	South	Technology	Machines	8749.95	5	0	2799.984
	Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19120	East	Technology	Copiers	8399.976	4	0.4	1119.9968
	Second Class	Consumer	United States	San Francisco	California	94122	West	Office Supplies	Supplies	8187.65	5	0	327.506
	Standard Class	Consumer	United States	San Antonio	Texas	78207	Central	Technology	Machines	8159.952	8	0.4	-1359.992
	Standard Class	Consumer	United States	New York City	New York	10024	East	Technology	Machines	6999.96	4	0	2239.9872
	Standard Class	Consumer	United States	Atlanta	Georgia	30318	South	Office Supplies	Binders	6354.95	5	0	3177.475
	Second Class	Consumer	United States	Jackson	Michigan	49201	Central	Office Supplies	Binders	5443.96	4	0	2504.2216
	Standard Class	Consumer	United States	Yonkers	New York	10701	East	Technology	Machines	4899.93	7	0	2400.9657
	Second Class	Consumer	United States	Philadelphia	Pennsylvania	19134	East	Office Supplies	Supplies	4663.736	7	0.2	-1049.3406
	Standard Class	Consumer	United States	Jamestown	New York	14701	East	Technology	Phones	4548.81	7	0	1228.1787
	Standard Class	Consumer	United States	Lancaster	Ohio	43130	East	Technology	Machines	4499.985	5	0.7	-6599.978
	Second Class	Consumer	United States	San Diego	California	92037	West	Technology	Machines	4476.8	4	0.2	503.64

SAMPLESUPERSTORE 18 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 37	23:36:16	SELECT * FROM SAMPLESUPERSTORE WHERE DISCOUNT>0.20 AND PROFIT<0	1348 row(s) returned	0.000 sec / 0.062 sec
✓ 38	23:40:08	SELECT * FROM SAMPLESUPERSTORE WHERE SEGMENT='CONSUMER' ORDER BY ...	5191 row(s) returned	0.046 sec / 0.032 sec

No object selected

Administration Schemas

Information

Result Grid

Form Editor

Field Types

Query Stats

Read On

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

SQL File 2* SQL File 4* SQL File 6* x samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

Don't Limit

SCHEMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Subcategory	Sales	Quantity	Discount	Profit
▶	Standard Class	Consumer	United States	Houston	Texas	77095	Central	Office Supplies	Appliances	0.444	1	0.8	-1.11
	Same Day	Corporate	United States	Waco	Texas	76706	Central	Office Supplies	Binders	0.556	1	0.8	-0.9452
	Standard Class	Corporate	United States	Chicago	Illinois	60623	Central	Office Supplies	Binders	0.836	1	0.8	-1.3376
	Same Day	Consumer	United States	Philadelphia	Pennsylvania	19120	East	Office Supplies	Binders	0.852	1	0.7	-0.5964
	Same Day	Corporate	United States	Mesquite	Texas	75150	Central	Office Supplies	Binders	0.876	1	0.8	-1.4016
	Standard Class	Corporate	United States	Huntsville	Texas	77340	Central	Office Supplies	Binders	0.898	1	0.8	-1.5715
	First Class	Consumer	United States	Houston	Texas	77041	Central	Office Supplies	Binders	0.984	2	0.8	-1.476
	First Class	Home Office	United States	San Francisco	California	94110	West	Technology	Accessories	0.99	1	0	0.4356
	Standard Class	Corporate	United States	Dallas	Texas	75217	Central	Office Supplies	Binders	1.044	1	0.8	-1.827
	Standard Class	Consumer	United States	Portland	Oregon	97206	West	Office Supplies	Binders	1.08	2	0.7	-0.792

SAMPLESUPERSTORE 19 x

Read On

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 38	23:40:08	SELECT * FROM SAMPLESUPERSTORE WHERE SEGMENT='CONSUMER' ORDER BY ...	5191 row(s) returned	0.046 sec / 0.032 sec
✓ 39	23:42:06	SELECT * FROM SAMPLESUPERSTORE ORDER BY SALES LIMIT 10	10 row(s) returned	0.031 sec / 0.000 sec

Object Info

The screenshot shows the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar shows the 'SCHEMAS' tab with a tree view containing 'bhartesh', 'sql_project' (expanded), 'Tables' (containing 'samplesuperstore'), 'Views', 'Stored Procedures', 'Functions', and 'sys'. The main editor window has a tab for 'SQL File 6*' and contains the following SQL query:

```
-- Question 8: Calculate the total Sales, total Quantity, total Discount, and total Profit for the entire dataset.
SELECT SUM(SALES) AS TOTAL_SALES,
       SUM(QUANTITY) AS TOTAL_QUANTITY,
       SUM(DISCOUNT) AS TOTAL_DISCOUNT,
       SUM(PROFIT) AS TOTAL_PROFIT
FROM SAMPLESUPERSTORE;
```

Below the query editor, the 'Result Grid' is displayed with the following data:

TOTAL_SALES	TOTAL_QUANTITY	TOTAL_DISCOUNT	TOTAL_PROFIT
2297200.860299955	37673	1561.0900000000954	286397.0217000013

At the bottom, the 'Output' tab is active, showing a table with columns: #, Time, Action, Message, and Duration / Fetch. The first row shows the execution of the query:

#	Time	Action	Message	Duration / Fetch
39	23:42:06	SELECT * FROM SAMPLESUPERSTORE ORDER BY SALES LIMIT 10	10 row(s) returned	0.031 sec / 0.000 sec
40	23:45:36	SELECT SUM(SALES) AS TOTAL_SALES, SUM(QUANTITY) AS TOTAL_QUANTITY, ...	1 row(s) returned	0.047 sec / 0.000 sec

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

Administration Schemas

No object selected

SQL File 6

samplesuperstore - Table

samplesuperstore - Table

samplesuperstore - Table

42

43 -- Question 9: Calculate the average Sales, average Discount, and average Profit for all transactions.

44 • SELECT AVG(SALES) AS AVERAGE_SALES,

45 AVG(DISCOUNT) AS AVERAGE_DISCOUNT,

46 AVG(PROFIT) AS AVERAGE_PROFIT

47 FROM SAMPLESUPERSTORE;

48

49

50

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	AVERAGE_SALES	AVERAGE_DISCOUNT	AVERAGE_PROFIT
▶	229.0580008304938	0.15620272163298934	28.656896307784802

Result 21

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 40	23:45:36	SELECT SUM(SALES) AS TOTAL_SALES, SUM(QUANTITY) AS TOTAL_QUANTITY, ...	1 row(s) returned	0.047 sec / 0.000 sec
✓ 41	23:46:25	SELECT AVG(SALES) AS AVERAGE_SALES, AVG(DISCOUNT) AS AVERAGE_DISC...	1 row(s) returned	0.031 sec / 0.000 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: bhartesh sql_project Tables samplesuperstore Views Stored Procedures Functions sys

SQL File 6* x samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

```
49 -- Question 10: Create a calculated column called Profit_Status using CASE.
50 SELECT SALES, PROFIT,
51 CASE
52 WHEN PROFIT > 0 THEN 'PROFITABLE'
53 WHEN PROFIT < 0 THEN 'LOSS'
54 ELSE 'NO PROFIT'
55 END AS 'PROFIT_STATUSES'
56 FROM SAMPLESUPERSTORE;
57
```

Result Grid

	SALES	PROFIT	PROFIT_STATUSES
▶	261.96	41.9136	PROFITABLE
	731.94	219.582	PROFITABLE
	14.62	6.8714	PROFITABLE
	957.5775	-383.031	LOSS
	22.368	2.5164	PROFITABLE
	48.86	14.1694	PROFITABLE
	7.28	1.9656	PROFITABLE
	907.152	90.7152	PROFITABLE
	18.504	5.7825	PROFITABLE
	114.9	34.47	PROFITABLE
	1705.184	85.3092	PROFITABLE
	911.424	68.3568	PROFITABLE

Result 22 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 41	23:40:25	SELECT AVG(SALES) AS AVERAGE_SALES, AVG(DISCOUNT) AS AVERAGE_DISC...	1 row(s) returned	0.031 sec / 0.000 sec
✓ 42	23:54:26	SELECT SALES, PROFIT, CASE WHEN PROFIT > 0 THEN 'PROFITABLE' WHEN PROFIT...	9994 row(s) returned	0.000 sec / 0.079 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Filter objects

- bhartesh
- sql_project
 - Tables
 - samplesuperstore
 - Views
 - Stored Procedures
 - Functions
- sys

SQL File 6* x samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

Don't Limit

```
74
75 -- Question 13: Find the top 5 Sub-Categories based on total Sales.
76 • SELECT SUBCATEGORY,
77       SUM(SALES) AS TOTAL_SALES
78 FROM SAMPLESUPERSTORE
79 GROUP BY SUBCATEGORY
80 ORDER BY TOTAL_SALES DESC
81 LIMIT 5;
82
```

Result Grid

	SUBCATEGORY	TOTAL_SALES
▶	Phones	330007.0540000001
	Chairs	328449.103000000076
	Storage	223843.60800000012
	Tables	206965.5320000001
	Dinners	203412.7330000001

Administration Schemas

Information

No object selected

Result 25 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 44	00:01:01	SELECT REGION, SUM(SALES) AS TOTAL_SALES, SUM(QUANTITY) AS TOTAL...	4 row(s) returned	0.063 sec / 0.000 sec
✓ 45	00:04:14	SELECT SUBCATEGORY, SUM(SALES) AS TOTAL_SALES FROM SAMPLESUPERS...	5 row(s) returned	0.078 sec / 0.000 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

SQL File 2* SQL File 4* SQL File 6* x samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

Don't Limit

```
82
83 -- Question 14: Find the average Discount for each Category and display only categories whose average Discount is greater than 15%.
84 SELECT CATEGORY,
85        AVG(DISCOUNT) AS AVERAGE_DISCOUNT
86 FROM SAMPLESUPERSTORE
87 GROUP BY CATEGORY
88 HAVING AVG(DISCOUNT) > 0.15;
89
90
```

Result Grid

	CATEGORY	AVERAGE_DISCOUNT
▶	Furniture	0.17392267798208297
	Office Supplies	0.15728509790906708

Administration Schemas

Information

No object selected

Result 26 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 45	00:04:14	SELECT SUBCATEGORY, SUM(SALES) AS TOTAL_SALES FROM SAMPLESUPERS...	5 row(s) returned	0.078 sec / 0.000 sec
✓ 46	00:07:43	SELECT CATEGORY, AVG(DISCOUNT) AS AVERAGE DISCOUNT FROM SAMPLES...	2 row(s) returned	0.093 sec / 0.000 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Filter objects

SCHEMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

SQL File 2* SQL File 4* SQL File 6* x samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

Don't Limit

```
89
90 -- Question 15: Find the cities whose total Sales are greater than the average city Sales using a subquery.
91 SELECT CITY, SUM(SALES) AS TOTAL_SALES FROM SAMPLESUPERSTORE
92 GROUP BY CITY
93 HAVING SUM(SALES) > (SELECT AVG(CITY_SALES) FROM (SELECT CITY, SUM(SALES) AS CITY_SALES
94 FROM SAMPLESUPERSTORE
95 GROUP BY CITY) AS CITY_TOTALS)
96 ORDER BY TOTAL_SALES DESC;
97
```

Result Grid

	CITY	TOTAL_SALES
▶	New York City	256368.161
	Los Angeles	175851.341
	Seattle	119540.742
	San Francisco	112669.09199999992
	Philadelphia	109077.01300000008
	Houston	64504.76039999994
	Chicago	48539.541000000034
	San Diego	47521.028999999995
	Jacksonville	44713.183
	Springfield	43054.342000000004
	Detroit	42446.94400000002
	Columbus	38706.242999999995

Result 28 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 48	00:18:37	SELECT CITY,SUM(SALES) AS TOTAL_SALES FROM SAMPLESUPERSTORE GROUP B...	92 row(s) returned	0.094 sec / 0.000 sec
✓ 49	00:19:07	SELECT CITY,SUM(SALES) AS TOTAL_SALES FROM SAMPLESUPERSTORE GROUP B...	92 row(s) returned	0.079 sec / 0.000 sec

No object selected

Administration Schemas

Information

Read O

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

SQL File 2* SQL File 4* SQL File 6* x samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

Don't Limit

98 -- Subqueries - 16 to 18

99 -- Question 16:Find all transactions where the Sales amount is greater than the overall average Sales.

100 • SELECT *

101 FROM SAMPLESUPERSTORE

102 WHERE SALES>(SELECT

103 AVG(SALES)

104 FROM SAMPLESUPERSTORE

105);

106

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Subcategory	Sales	Quantity	Discount	Profit
▶	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	261.96	2	0	41.9136
	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.94	3	0	219.582
	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	957.5775	5	0.45	-383.031
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	907.152	6	0.2	90.7152
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Tables	1706.184	9	0.2	85.3092
	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Phones	911.424	4	0.2	68.3568
	Standard Class	Consumer	United States	Seattle	Washington	98103	West	Office Supplies	Binders	407.976	3	0.2	132.5922
	Standard Class	Consumer	United States	Madison	Wisconsin	53711	Central	Office Supplies	Storage	665.88	6	0	13.3176
	Standard Class	Consumer	United States	Orem	Utah	84057	West	Furniture	Tables	1044.63	3	0	240.2649
	Standard Class	Consumer	United States	Philadelphia	Pennsylvania	19140	East	Furniture	Bookcases	3083.43	7	0.5	-1665.0522
	First Class	Corporate	United States	Richardson	Texas	75080	Central	Technology	Phones	1097.544	7	0.2	123.4737
	Standard Class	Consumer	United States	Houston	Texas	77041	Central	Furniture	Bookcases	523.3002	2	0.22	46.0764

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 49	00:19:07	SELECT CITY,SUM(SALES) AS TOTAL_SALES FROM SAMPLESUPERSTORE GROUP B...	92 row(s) returned	0.079 sec / 0.000 sec
✓ 50	00:22:25	SELECT * FROM SAMPLESUPERSTORE WHERE SALES>(SELECT AVG(SALES...	2360 row(s) returned	0.031 sec / 0.031 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: bhartesh sql_project Tables samplesuperstore Views Stored Procedures Functions sys

SQL File 2* SQL File 4* SQL File 6* samplesuperstore - Table samplesuperstore - Table samplesuperstore - Table

Don't Limit

```
107 -- Question 17: Find the Sub-Category with the highest total Profit using a subquery.
108 SELECT SUBCATEGORY, SUM(PROFIT) AS SUBCATEGORY_TOTAL
109 FROM SAMPLESUPERSTORE
110 GROUP BY SUBCATEGORY
111 HAVING SUM(PROFIT) = (SELECT MAX(SUBCATEGORY_TOTAL)
112 FROM (SELECT SUBCATEGORY, SUM(PROFIT) AS SUBCATEGORY_TOTAL
113 FROM SAMPLESUPERSTORE
114 GROUP BY SUBCATEGORY) AS SUBCATEGORY_TOTAL);
115
```

Result Grid

	SUBCATEGORY	SUBCATEGORY_TOTAL
1	Copiers	55517.824900000001

Administration Schemas Information

No object selected

Result: 39 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
53	00:34:16	SELECT SUBCATEGORY,SUM(PROFIT) AS SUBCATEGORY_TOTAL FROM SAMPLESU...	Error Code: 1248. Every derived table must have its own alias	0.000 sec
54	00:35:26	SELECT SUBCATEGORY,SUM(PROFIT) AS SUBCATEGORY_TOTAL FROM SAMPLESU...	1 row(s) returned	0.062 sec / 0.000 sec

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigation

SCHEMAS

Filter objects

bhatesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

Bhatesh*

SQL File 2*

SQL File 4*

SQL File 6*

Don't Limit

SQL Editor

```
116 -- Question 18: Find all States whose total Sales are greater than the average total Sales of all States.
117 SELECT STATE, SUM(SALE) AS TOTAL_STATE
118 FROM SAMPLESUPERSTORE
119 GROUP BY STATE
120 HAVING SUM(SALE) > (SELECT AVG(SALE)
121 FROM (SELECT STATE, SUM(SALE) AS STATE_SALE
122 FROM SAMPLESUPERSTORE
123 GROUP BY STATE) AS STATE_TOTALS)
124
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

STATE	TOTAL_STATE
Kentucky	0
California	0
Florida	0
North Carolina	0
Washington	0
Texas	0
Wisconsin	0
Utah	0
Nebraska	0
Pennsylvania	0
Illinois	0
Minnesota	0

Administration

Schemas

Information

No object selected

Result Grid

Form Editor

Field Types

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	09:57:26	USE SQL_PROJECT	0 row(s) affected	0.000 sec
2	09:57:30	SELECT STATE, SUM(SALE) AS TOTAL_STATE FROM SAMPLESUPERSTORE GROUP BY STATE	49 row(s) returned	0.219 sec / 0.000 sec

Local instance MySQL80 X

File Edit View Query Database Server Tools Scripting Help

Navigator: www.mysql.com/doc/relnotes/mysql/8.0/en/whats-new-in-8.0.html

SCHMAS

Filter objects

- bhartesh
 - sql_project
 - Tables
 - samplesuperstore
 - Views
 - Stored Procedures
 - Functions
 - sys

SQL File 2* SQL File 4* SQL File 6* x

Don't Limit

```
125 -- Joins - 19 to 21
126 -- Question 19: Using a JOIN, display each Category along with its total Sales and total Profit.
127 SELECT S.CATEGORY,S.TOTAL_SALES,P.TOTAL_PROFIT
128 FROM (SELECT CATEGORY,SUM(SALES) AS TOTAL_SALES
129 FROM SAMPLESUPERSTORE
130 GROUP BY CATEGORY) AS S
131 JOIN (SELECT CATEGORY,SUM(PROFIT) AS TOTAL_PROFIT
132 FROM SAMPLESUPERSTORE
133 GROUP BY CATEGORY) AS P
134 ON S.CATEGORY=P.CATEGORY;
135
```

Result Grid

	CATEGORY	TOTAL_SALES	TOTAL_PROFIT
▶	Furniture	741999.7952999998	18451.2728
	Office Supplies	719047.0320000029	122490.80080000011
	Technology	836154.0329999966	145454.94809999999

Administration Schemas

Information

No object selected

Result 2 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 2	09:57:30	SELECT STATE,SUM(STATE) AS TOTAL_STATE FROM SAMPLESUPERSTORE GROU...	45 row(s) returned	0.219 sec / 0.000 sec
✓ 3	10:06:34	SELECT S.CATEGORY,S.TOTAL_SALES,P.TOTAL_PROFIT FROM (SELECT CATEGORY...	3 row(s) returned	0.125 sec / 0.000 sec

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

Bhartesh* SQL File 2* SQL File 4* SQL File 6*

Don't Limit

137

SELECT R.REGION,R.TOTAL_SALES,T.OVERALL_SALES,

138 ROUND((R.TOTAL_SALES/T.OVERALL_SALES)*100,2) AS PERCENTAGE_SALES

139 FROM (SELECT REGION,SUM(SALES) AS TOTAL_SALES

140 FROM SAMPLESUPERSTORE

141 GROUP BY REGION

142) AS R

143 JOIN

144 (SELECT REGION,SUM(SALES) AS OVERALL_SALES

145 FROM SAMPLESUPERSTORE

146 GROUP BY REGION

147) AS T;

148

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	REGION	TOTAL_SALES	OVERALL_SALES	PERCENTAGE_SALES
▶	East	678781.2399999979	391721.9050000003	173.28
	Central	501239.8908000005	391721.9050000003	127.96
	West	725457.8245000006	391721.9050000003	185.2
	South	391721.9050000003	391721.9050000003	100
	East	678781.2399999979	725457.8245000006	93.57
	Central	501239.8908000005	725457.8245000006	69.09
	West	725457.8245000006	725457.8245000006	100
	South	391721.9050000003	725457.8245000006	54
Fact	East	678781.2399999979	501739.8908000005	135.47

Result 3

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✖	5 10:15:42	SELECT R.REGION,R.TOTAL_SALES,T.OVERALL_SALES, ROUND((R.TOTAL_SALES/...	Error Code: 1140. In aggregated query without GROUP BY, expression #1 of SELECT list co...	0.000 sec
✔	6 10:16:09	SELECT R.REGION,R.TOTAL_SALES,T.OVERALL_SALES, ROUND((R.TOTAL_SALES/...	16 row(s) returned	0.094 sec / 0.000 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

SQL File 6*

Don't Limit

149 -- Question 21: Using a JOIN, identify the Product Types that have both total sales greater than 100,000 and total Profit greater than 10,000.

150 SELECT S.SUBCATEGORY, S.TOTAL_SALES, T.TOTAL_PROFIT

151 FROM (SELECT SUBCATEGORY, SUM(SALES) AS TOTAL_SALES

152 FROM SAMPLESUPERSTORE

153 GROUP BY SUBCATEGORY

154) AS S

155 JOIN

156 (SELECT SUBCATEGORY, SUM(PROFIT) AS TOTAL_PROFIT

157 FROM SAMPLESUPERSTORE

158 GROUP BY SUBCATEGORY

159) AS T

160 ON S.SUBCATEGORY=T.SUBCATEGORY

161 WHERE S.TOTAL_SALES>100000

162 AND T.TOTAL_PROFIT>10000

163 ORDER BY S.TOTAL_SALES DESC;

...

Result Grid

Filter Rows:

Export:

Wrap Cell Content: 16

	SUBCATEGORY	TOTAL_SALES	TOTAL_PROFIT
▶	Phones	330007.0540000001	44515.7306
	Chairs	328449.10300000076	26590.166300000026
	Storage	223843.60800000012	21278.826399999998
	Binders	203412.7330000001	30221.763299999995
	Accessories	167380.3180000001	41936.63569999993
	Copiers	149528.02999999994	55617.82490000001

Result: 4 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
❌ 11	10:27:56	SELECT S.'Product Type', S.Total_Sales, P.Total_Profit FROM (SELECT ...	Error Code: 1146. Table 'sql_project.superstore' doesn't exist	0.016 sec
✅ 12	10:28:50	SELECT S.SUBCATEGORY, S.TOTAL_SALES, T.TOTAL_PROFIT FROM (SELECT SUBCA...	7 row(s) returned	0.078 sec / 0.000 sec

Administration Schemas

Information

No object selected

Result Grid

Form Editor

Read On

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

SQL File 2* SQL File 4* SQL File 6*

Don't Limit

```
165 -- Set Operators - 22 to 23
166 -- Question 22:Using UNION, create a single list containing all Cities and States from the dataset, without duplicate values.
167 SELECT CITY AS LOCATION
168 FROM SAMPLESUPERSTORE
169 UNION
170 SELECT STATE AS LOCATION
171 FROM SAMPLESUPERSTORE;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

LOCATION
Henderson
Los Angeles
Fort Lauderdale
Concord
Seattle
Fort Worth
Madison
West Jordan
San Francisco
Fremont
Philadelphia
Orem
Houston
Richardson
Menomville

Administration Schemas

Information

No object selected

Result 5

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	10:28:50	SELECT S.SUBCATEGORY,S.TOTAL_SALES,T.TOTAL_PROFIT FROM (SELECT SUBCA...	7 row(s) returned	0.078 sec / 0.000 sec
13	10:32:27	SELECT CITY AS LOCATION FROM SAMPLESUPERSTORE UNION SELECT STATE AS...	579 row(s) returned	0.047 sec / 0.000 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

SQL File 2* SQL File 4* SQL File 6* x

Don't Limit

173 -- Question 23: Using a set operator such as INTERSECT, find the Cities that appear in both the Consumer and Corporate segments.
174 SELECT CITY
175 FROM SAMPLESUPERSTORE
176 WHERE SEGMENT='CONSUMER'
177 INTERSECT
178 SELECT CITY
179 FROM SAMPLESUPERSTORE
180 WHERE SEGMENT='CORPORATE';
181

Result Grid

CITY
Henderson
Fort Lauderdale
Los Angeles
Concord
Seattle
San Francisco
Philadelphia
Orem
Dover
Troy
Springfield
New York City
Jackson

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 13	10:32:27	SELECT CITY AS LOCATION FROM SAMPLESUPERSTORE UNION SELECT STATE AS ...	579 row(s) returned	0.047 sec / 0.000 sec
✓ 14	10:36:21	SELECT CITY FROM SAMPLESUPERSTORE WHERE SEGMENT='CONSUMER' INTERS...	260 row(s) returned	0.062 sec / 0.000 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

bhantesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

SQL File 2* SQL File 4* SQL File 6* x

Don't Limit

```
181
182 -- Question 24: Use the ROW_NUMBER() window function to assign a row number to each transaction based on Sales in descending order. Display Sales, Profit, and Row Number.
183 • SELECT SALES, PROFIT,
184 ROW_NUMBER () OVER (ORDER BY SALES DESC)
185 FROM SAMPLESUPERSTORE;
186
187
188
189
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	SALES	PROFIT	ROW_NUMBER () OVER (ORDER BY SALES DESC)
▶	22638.48	-1811.0784	1
	17499.95	8399.976	2
	13999.96	6719.9808	3
	11199.968	3919.9888	4
	10499.97	5039.9856	5
	9992.74	4946.37	6
	9449.95	4630.4755	7
	9099.93	2365.9818	8
	8749.95	2799.984	9
	8399.976	1119.9968	10
	8187.65	327.506	11
	8159.952	-1359.992	12

Result 8 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓	15 10:38:35	SELECT CITY FROM SAMPLESUPERSTORE WHERE SEGMENT=CONSUMER INTER...	260 row(s) returned	0.093 sec / 0.000 sec
✓	16 10:43:31	SELECT SALES, PROFIT, ROW_NUMBER () OVER (ORDER BY SALES DESC) FROM SA...	9994 row(s) returned	0.031 sec / 0.016 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

bhartesh

sql_project

Tables

samplesuperstore

Views

Stored Procedures

Functions

sys

SQL File 2* SQL File 4* SQL File 6* x

Don't Limit

```
186
187 -- Question 25: Use the RANK() window function to rank transactions based on Profit in descending order. Display Sales, Profit, and the Profit Rank.
188 • SELECT SALES, PROFIT,
189 RANK() OVER (ORDER BY PROFIT DESC)
190 FROM SAMPLESUPERSTORE;
191
192
193
194
```

Result Grid

Filter Rows: Export: Wrap Cell Contents: Fetch rows:

	SALES	PROFIT	RANK() OVER (ORDER BY PROFIT DESC)
▶	17499.95	8399.976	1
	13999.96	6719.9808	2
	10499.97	5039.9856	3
	9892.74	4946.37	4
	9449.95	4630.4755	5
	11199.968	3919.9888	6
	6354.95	3177.475	7
	8749.95	2799.984	8
	5399.91	2591.9568	9
	5443.96	2504.2216	10
	4899.93	2400.9657	11
	9099.93	2365.9818	12

Result 9 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 16	10:43:31	SELECT SALES, PROFIT, ROW_NUMBER () OVER (ORDER BY SALES DESC) FROM SA...	5994 row(s) returned	0.031 sec / 0.016 sec
✓ 17	10:45:19	SELECT SALES, PROFIT, RANK() OVER (ORDER BY PROFIT DESC) FROM SAMPLESUP...	5994 row(s) returned	0.047 sec / 0.031 sec